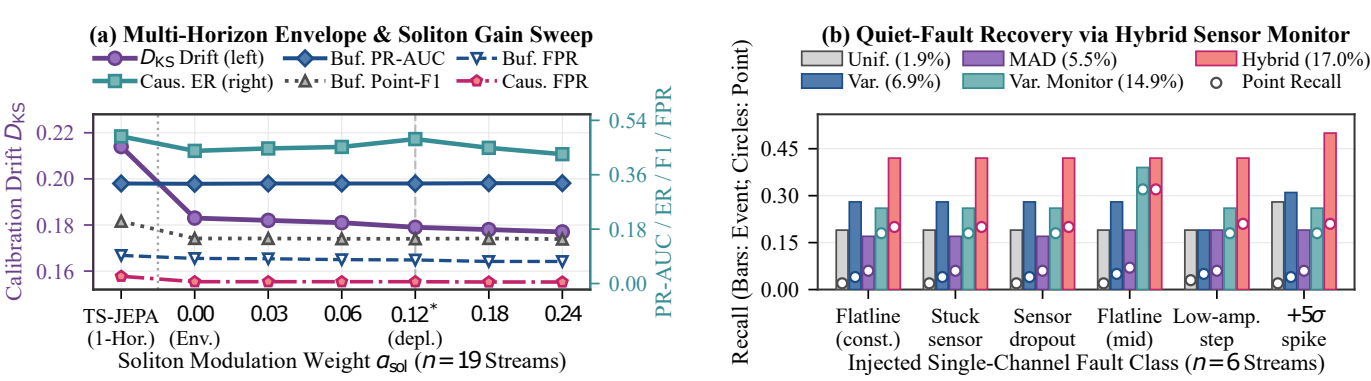

(a) Multi-Horizon Envelope & Soliton Gain Sweep
D_{KS} Drift (left)
Buf. PR-AUC
Buf. FPR
Caus. ER (right)
Buf. Point-F1
Caus. FPR
Calibration Drift D_{KS}
PR-AUC / ER / F1 / FPR
0.22
0.20
0.18
0.16
0.54
0.36
0.18
0.00
TS-JEPA (1-Hor.)
0.00 (Env.)
0.03
0.06
0.12* (depl.)
0.18
0.24
Soliton Modulation Weight a_{sol} ($n = 19$ Streams)
(b) Quiet-Fault Recovery via Hybrid Sensor Monitor
Unif. (1.9%)
MAD (5.5%)
Hybrid (17.0%)
Var. (6.9%)
Var. Monitor (14.9%)
Point Recall
Recall (Bars: Event; Circles: Point)
0.45
0.30
0.15
0.00
Flatline (const.)
Stuck sensor
Sensor dropout
Flatline (mid)
Low-amp. step
+5σ spike
Injected Single-Channel Fault Class ($n = 6$ Streams)